DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISHAL

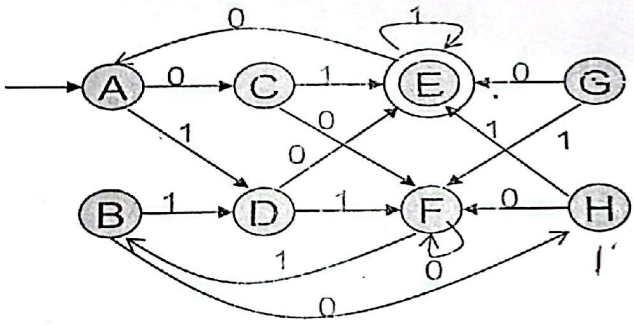
FINAL EXAMINATION
Course Title: Theory of Computation
Course Code: CSE-3203
3rd Year, 2nd Semester, Session: 2021-22

Time: 3 hours **Marks:** 60

Answer any Five (5) Questions from the followings.

1. a) A hierarchy of classes of formal languages has been classified as The Chomsky Hierarchy. Describe with suitable figure(s). 4
   b) Describe the applications of Finite Automata. 4
   c) Describe about the context-free grammer with proper example(s). 4

2. a) Compute the equivalent states in a DFA by using Table-filling algorithm for the following: 7



   b) Show that the grammar ({S}, {a, b}, R, S) with rules R = S → aS | aSbS | ϵ is ambiguous. 3
   c) Does a push down automata have memory? Justify. 2

3. a) Describe the Turing Machine with example. Give advantages of it. 3
   b) Construct a Turing Machine that accepts the language of palindromes over {a,b} . Also specify the moves to trace the strings abaa,abba,aabaa. 6
   c) What is meant by a halting problem in a Turing Machine? Explain with an example. 3

4. a) Let, Σ = {0, 1}
   Construct a DFA for the following language:
   L = {w | w is a binary string that has even number of 1s and even number of 0s} 6
   b) Construct a NFA for the following: Strings where the first symbol is present somewhere later on at least once. 6

5. a) Describe the Advantages & Limitations for NFA. 3
   c) Build an NFA for the following language: L = { w | w ends in 011} 4
   d) Convert the NFA from Question 5(c), to DFA. 5

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6. a) Describe the importance of explicit ε-transitions in finite automata. 2

b) Build an ε-NFA for the following language: 4

L = {w | w is empty, or if non-empty will end in 01}

c) Convert ε-NFA to DFA based on Question 6(b) 6

7. a) Describe the "bad case" where states(DFA) >> states(NFA). Amalyze it with the Pigeon hole principle. 5

b) Describe the interchangeable relation between Regular Expressions (RE) and Finite Automata with figure. 3

c) Convert the following RE to ε-NFA: (0+1)*01(0+1)* 4

8. a) Analyze the terms: Theorems, Lemmas and Corollaries. 3

b) Give pushdown automata that recognize the following languages: 4

(a) A = { w ∈ {0, 1} * | w contains at least three 1s }

(b) B = { w ∈ {0, 1} * | w = w^R and the length of w is odd }

c) Use the pumping lemma to prove that the language L = { 0^i | i is a perfect square} is not regular 5

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